Unit 2

TECHNOLOGY AND INCENTIVES

OUTLINE

A. Introduction
B. Topic 1: Economic incentives
C. Topic 2: Principles of economic modelling
D. Topic 3: Gains from specialisation and trade
E. Topic 4: Modelling a firm’s choice of technology

A. Introduction

The Context for This Unit

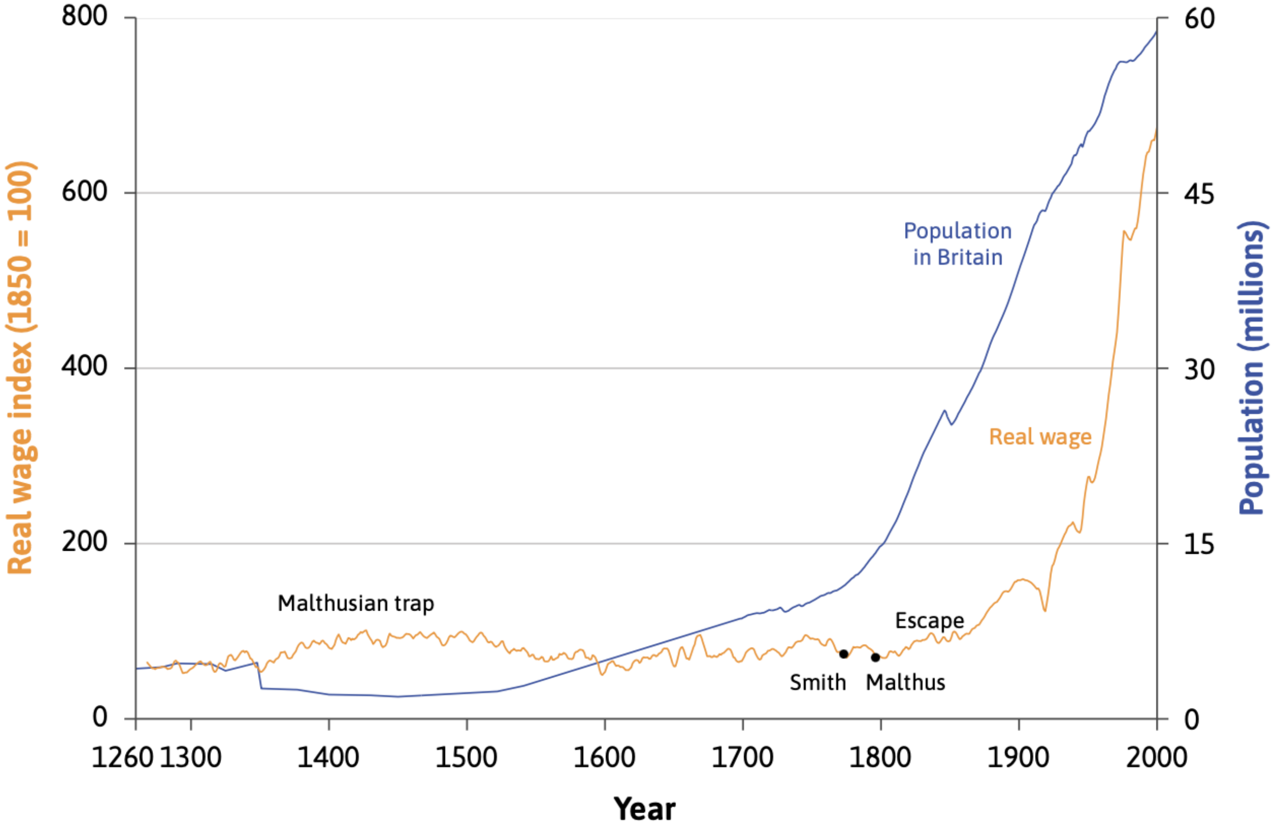
The recent rapid, sustained increase in income and living standards is largely due to technological progress.

(Unit 1)

However, these major changes started very suddenly, 200 years ago.

- How did the technological revolution start?
- Why did it not start earlier?

This Unit



Use economic models to explain the rapid growth in real wages and population in the last 2 centuries.

B. Decision making

A model of decision making

- **Opportunity cost** – the value of the next best action not taken
- **Reservation option** – the next best alternative
- **Economic cost** = direct costs incurred by taking an action + opportunity cost
- **Economic rent** = net benefit from option taken – opportunity cost

Scenario: Are you going to the concert?

- A ticket costs \$25
- Enjoyment value of concert is \$55
- Net benefit of concert = \$55 - \$25 = \$30

- Reservation option is earning \$22 for babysitting
- Opportunity cost = \$22

- Economic cost of concert = \$25 + \$22 = \$47
- Economic rent = \$55 - \$47 = \$30 - \$22 = \$8

Identifying opportunity costs

- Use your Gen AI / LLM assistant to:

1. *Prompt*:

"Provide a scenario where a student must choose between attending university or starting a full-time job after high school. Include the benefits and costs associated with each option."

2. *Prompt*:

"Provide a scenario where someone decides between buying a new smartphone or investing that money in a savings account with interest. Describe the potential outcomes of both choices."

-> Identify the opportunity cost in each scenario.

Calculating opportunity costs & Economic rent

EXERCISE 2.1: Suppose you have the choice of going to a theatre concert (option A) with a $15 admission cost, or a park concert (option B), which is free and near the theatre but happens at the same time. Use this information to help you fill in the table below.

	A high value on the theatre choice (A)	**A low value on the theatre choice (A)**
Out-of-pocket cost (price of ticket for A)	$25	$25
Enjoyment of theatre concert (A)	$50	$35
Opportunity cost (foregone pleasure of B, the park concert)	$15	$15
Economic cost		
Economic rent		
Decision (which concert would you go to?)		

Calculating opportunity costs & Economic rent (solution)

EXERCISE 2.1: Suppose you have the choice of going to a theatre concert (option A) with a $15 admission cost, or a park concert (option B), which is free and near the theatre but happens at the same time. Use this information to help you fill in the table below.

	A high value on the theatre choice (A)	**A low value on the theatre choice (A)**
Out-of-pocket cost (price of ticket for A)	$25	$25
Enjoyment of theatre concert (A)	$50	$35
Opportunity cost (foregone pleasure of B, the park concert)	$15	$15
Economic cost	A) 25 + 15 B) 0 + (50 – 25)	A) 25 + 15 B) 0 + (35 – 25)
Economic rent	A) 50 – 40 B) 15 - 25	A) 35 – 40 B) 15 - 10
Decision (which concert would you go to?)	Theatre	Park

Innovation rent and relative prices

- **Innovation rents** – a form of economic rent. The extra profits made by exploiting an invention. Provide **incentives** for taking action.

- **Relative prices** – the price of one option relative to another, often expressed as a ratio of the two prices. An important factor in determining economic **incentives**.

C. Topic 2: Principles of economic modelling

Model-building

When we build a model, the process follows these steps:

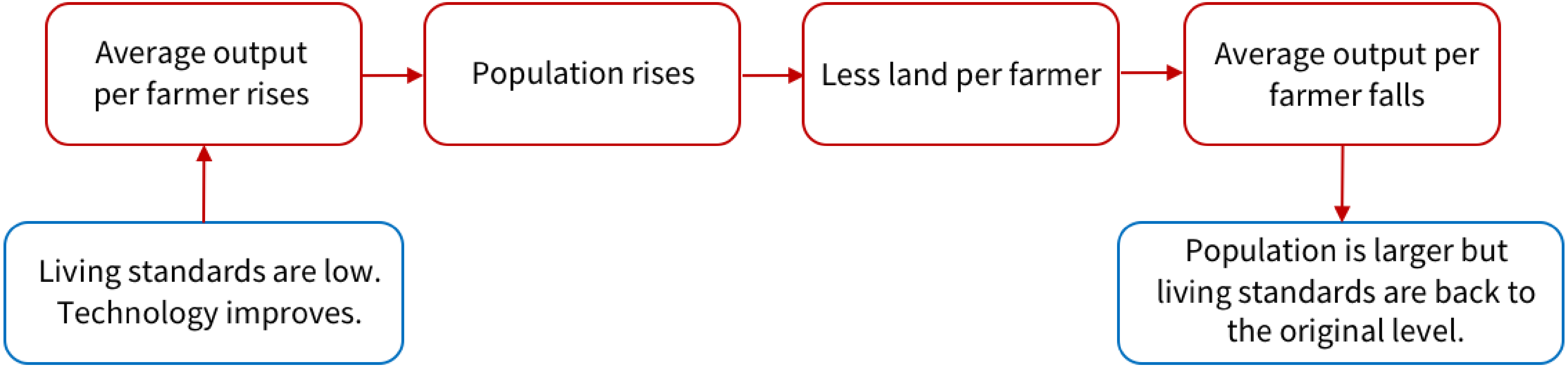
1. We begin with a well-defined question.
2. We construct a simplified description of the conditions under which people take actions.
3. We describe in simple terms what determines the actions that people take.
4. We determine how each of their actions affects each other.
5. We determine the outcome of these actions. This is often an equilibrium (something is constant).
6. Finally, we try to get more insight by studying what happens to certain variables when conditions change.

Key ideas in a model

- **Equilibrium** – a situation that is self-perpetuating, meaning that there is no tendency for the situation to change unless an external force for change is introduced
- **Endogenous variables** – variables whose values are determined by relationships built into the mode
- **Exogenous variables** – variables whose values are determined outside the model
- **Ceteris paribus** – holding other things equal

Example: The Malthusian model

Why technological improvement in farming doesn't raise living standards



Principles of economic modelling

- In groups, interpret the following statement (from the statistician George Box): "*All models are wrong, but some are useful*". Come up with a list of criteria for a "useful" model.
- Use an AI assistant to generate criteria for a 'useful' economic model by interpreting George Box's statement. Example Prompt:
  - "Interpret the statement 'All models are wrong, but some are useful' by George Box. List and explain criteria that make an economic model useful."
- Compare the AI's criteria with your own to understand the principles of economic modeling.

Exercise: Building a model

- In groups, apply the 6 steps of the “Model-building” box to build a model of the market for umbrellas.
- Exercise 2.10: Suppose you build the model in which the predicted number of umbrellas sold by a shop depends on their colour and price, ceteris paribus.
  - The colour and the price are the variables that are used to predict sales. Which other variables are being held constant?
  - Which of the following questions do you think this model might be able to answer? In each case, suggest improvements to the model that might help you to answer the question.
    - Why are annual umbrella sales higher in the capital city than in other towns?
    - Why are annual umbrella sales higher in some shops in the capital city than others?
    - Why have weekly umbrella sales in the capital city risen over the last six months?

Importance of assumptions

- In groups, read this article from The Economist.
- Explain why economic models did not predict the global financial crisis (what assumptions were incorrect?).

Microsoft Word Document

D. Topic 3: Gains from specialisation and trade

Specialization

Specialization means each person does just one or a few of the large number of tasks necessary for some project or goal.

Specialization can happen for three reasons:

- Learning by doing
- Difference in ability
- **Economies of scale**

Comparative and absolute advantage

	Production if 100% of time is spent on one good
Greta	1,250 apples or 50 tonnes of wheat
Carlos	1,000 apples or 20 tonnes of wheat

- Being more productive in producing a good means having an **absolute advantage.**
- Having a lower relative cost of producing a good means having a **comparative advantage.**

Model and intuition

- Ricardian trade model (2 people, 2 goods) to illustrate how specialization and trade can be mutually beneficial.
- The link between comparative advantage and opportunity cost (i.e., comparative advantage is determined by opportunity cost).
- Specialization and trade patterns depend on comparative advantage and not absolute advantage.

Example

- Imagine Country A can produce more of both wheat and cloth than Country B.
  - *Absolute Advantage*: Country A has an absolute advantage in both wheat and cloth production.
  - *Comparative Advantage*: However, if Country B has a lower opportunity cost in producing cloth (meaning it gives up less wheat to produce cloth), then Country B has a comparative advantage in cloth.
  - *Trade Pattern*: Country A should specialize in wheat, and Country B should specialize in cloth. Both countries will then be better off by trading, as B produces cloth at lower opportunity cost, while A benefits by importing cloth and producing wheat.

Comparative and absolute advantage

	Opportunity cost of 1 tonne of wheat	Opportunity cost of 1 apple
Greta	25 apples	0.04 tonnes of wheat
Carlos	50 apples	0.02 tonnes of wheat

- Being more productive in producing a good means having an **absolute advantage.**
- Having a lower relative cost of producing a good means having a **comparative advantage.**

Self-sufficiency vs specialization and trade

In self-sufficiency:

- Greta uses 40% of her time on apple production and 60% on wheat
- Carlos spends 30% on apple production and 70% on wheat

		Self-sufficiency	Complete specialization and trade		
			Production	Consumption	Trade
		(1)	(2)	(3)	(4)
Greta	Apples	500	0	600	Buys 600 apples
	Wheat	30	50	35	Sells 15t wheat
Carlos	Apples	300	1,000	400	Sells 600 apples
	Wheat	14	0	15	Buys 15t wheat
Total	Apples	800	1,000	1,000	
	Wheat	44	50	50	

Sources of comparative advantage

- In groups, brainstorm the different sources of comparative advantage (e.g. natural resources, human capital).
- Guess the country.

ECI (TRADE)	ECI (TECHNOLOGY)	ECI (RESEARCH)	PRODUCT EXPORTS	PRODUCT IMPORTS	EXPORTS PER CAPITA	IMPORTS PER CAPITA	EXPORTS GROWTH
4th	32nd	63rd	$645B	$481B	$12.5k	$9.3k	-10.7%
Out of 132	Out of 96	Out of 137	Ranking 6 / 226	Ranking 13 / 226	Ranking 35 / 209	Ranking 57 / 209	-$77.2B
(ECI: 1.83, 2023)	(ECI: 0.73, 2021)	(ECI: -0.13, 2023)	(2023)	(2023)	(2023)	(2023)	(2022-2023)

The Economic Complexity Index (ECI) and the Product Complexity Index (PCI) are, respectively, measures of the relative knowledge intensity of an economy or a product.

Guess the country

- Play the game on your own and discuss your reasoning.

[TRADLE]

E. Topic 4: Modelling a firm’s choice of technology

Intuition

- In these sections we build a model of technology choice.
- Technologies are defined according to their energy-to-labour ratio and the firm chooses the lowest-cost technology.
- This model is used to explain how creative destruction occurs and, later, explain the role of incentives (relative prices) in prompting the Industrial Revolution in the UK.

Firms, technology and production

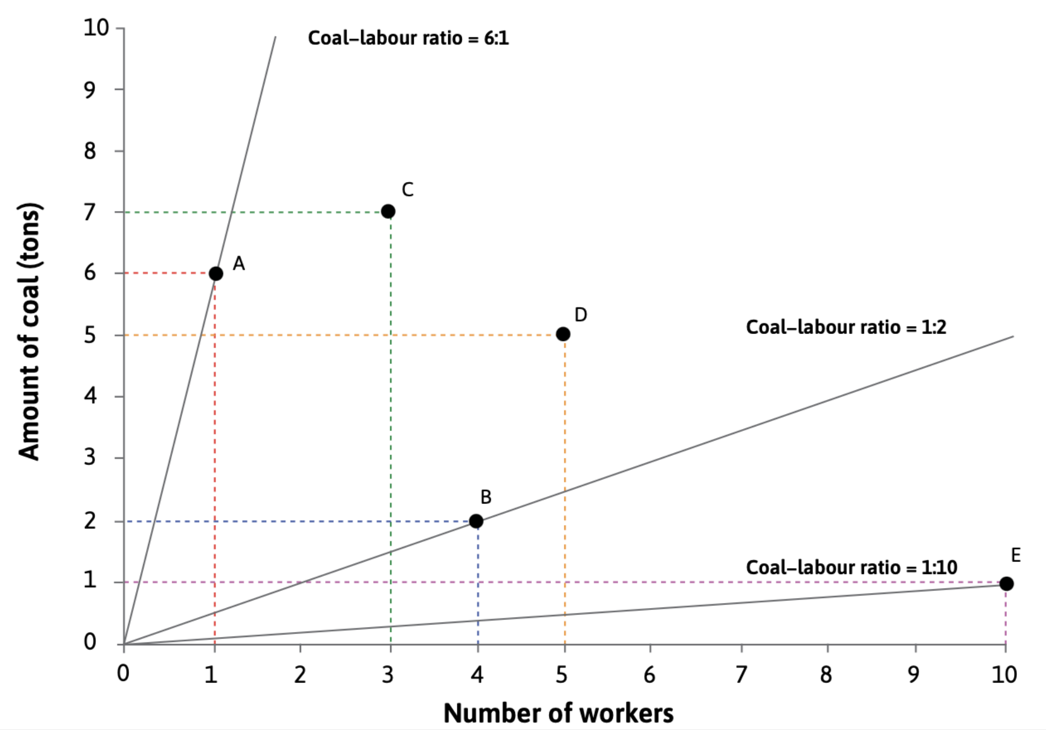
- **Division of labour** – specialisation in producing things either within a firm or across society
- **Production technology** - the process that a firm uses to convert a set of inputs into an output that it can sell.
- **Factors of production** – the inputs into the production process (e.g. raw materials, labour, capital goods, energy)
- **Production function** – a relationship that tell us how much output will be produced given the amounts of inputs used

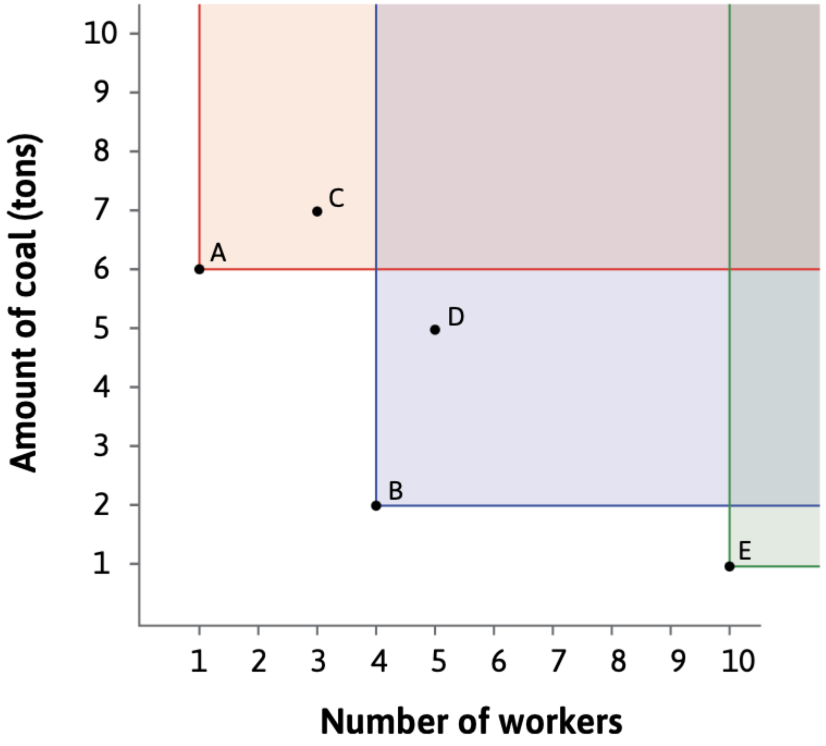
Fixed-proportions technology and CRS

Number of machines, M	Number of workers, N	Amount of energy, E (kWh)	Output, Y (litres)
3	1	80	50
6	2	160	100
9	3	240	150
12	4	320	200

Which technology?

Suppose there are 5 possible technologies for producing 100 metres of cloth. Which one should a firm choose?



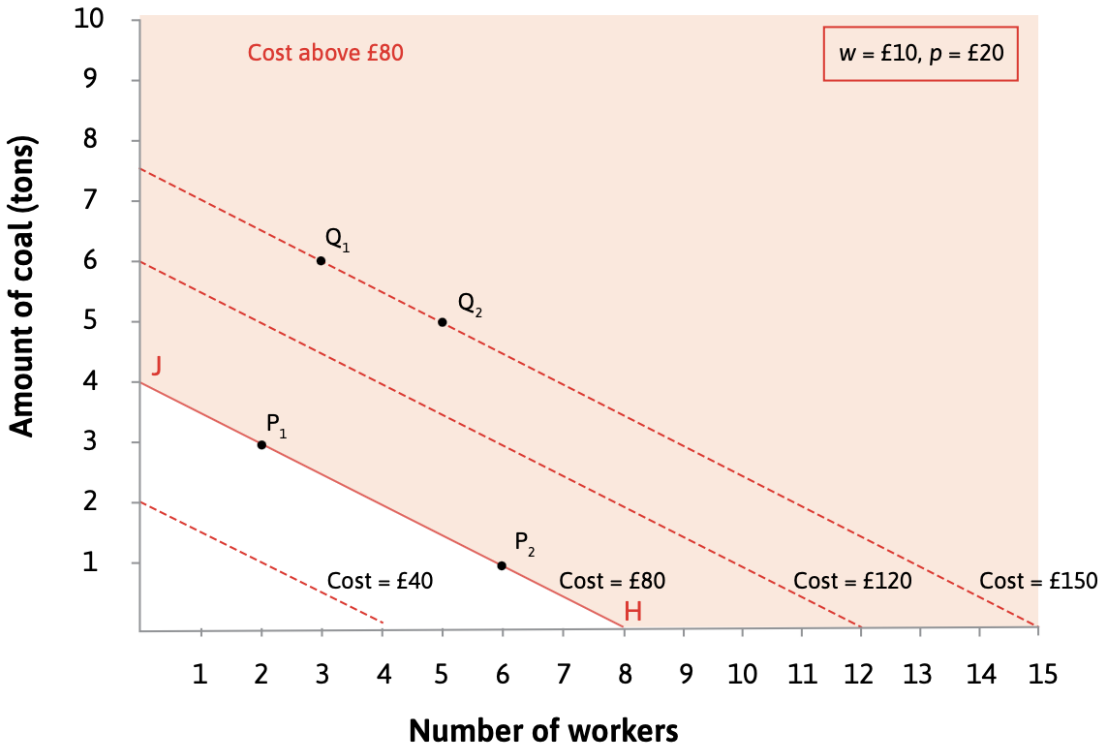


Which technology?

- Assume the motive of the firm is to maximise profit. This means they will want to minimise the cost. Also, assume w = £10 and p = £20.

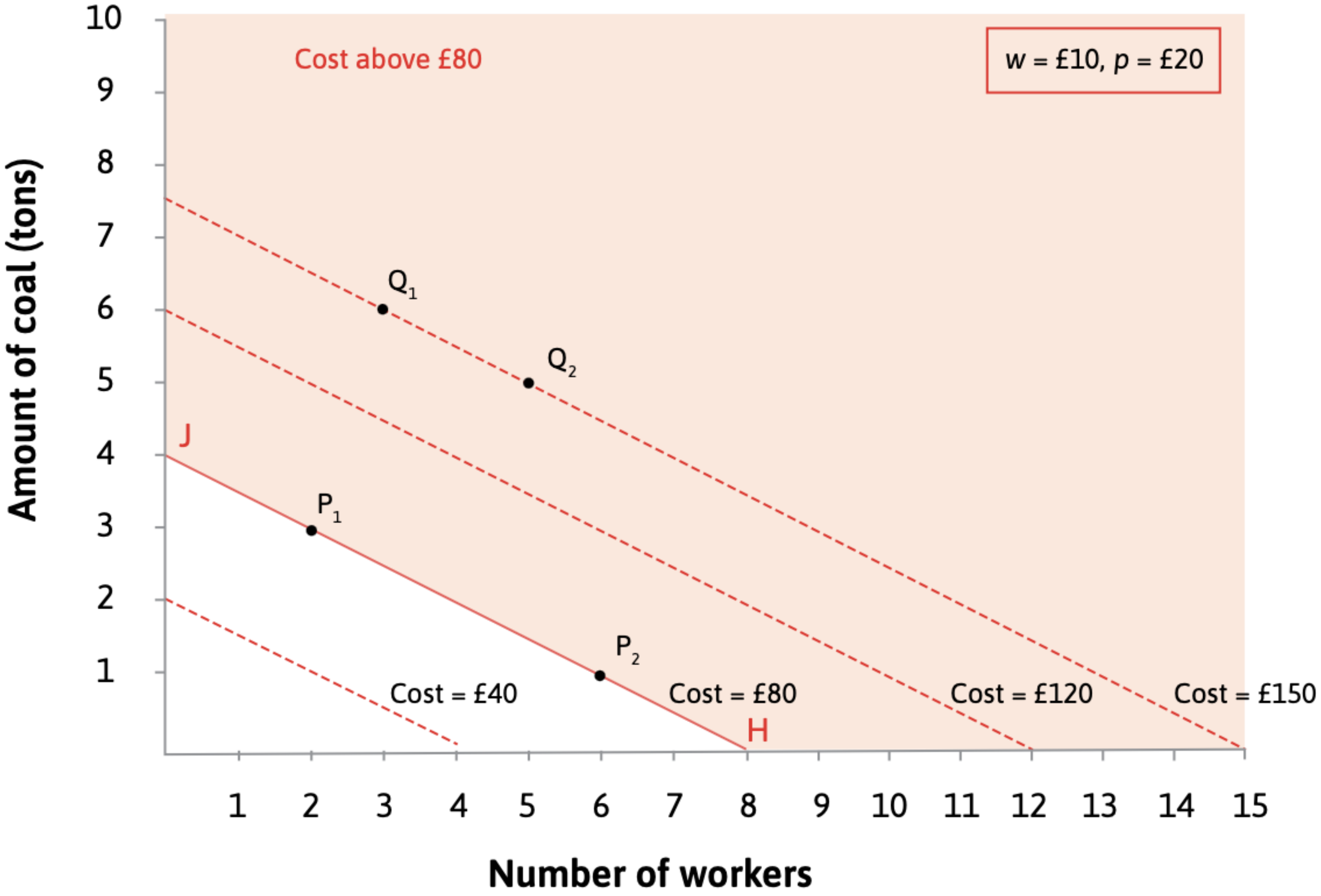
Cost = (wage × workers) + (price of a ton of coal × number of tons)

$$= (w \times N) + (p \times R)$$

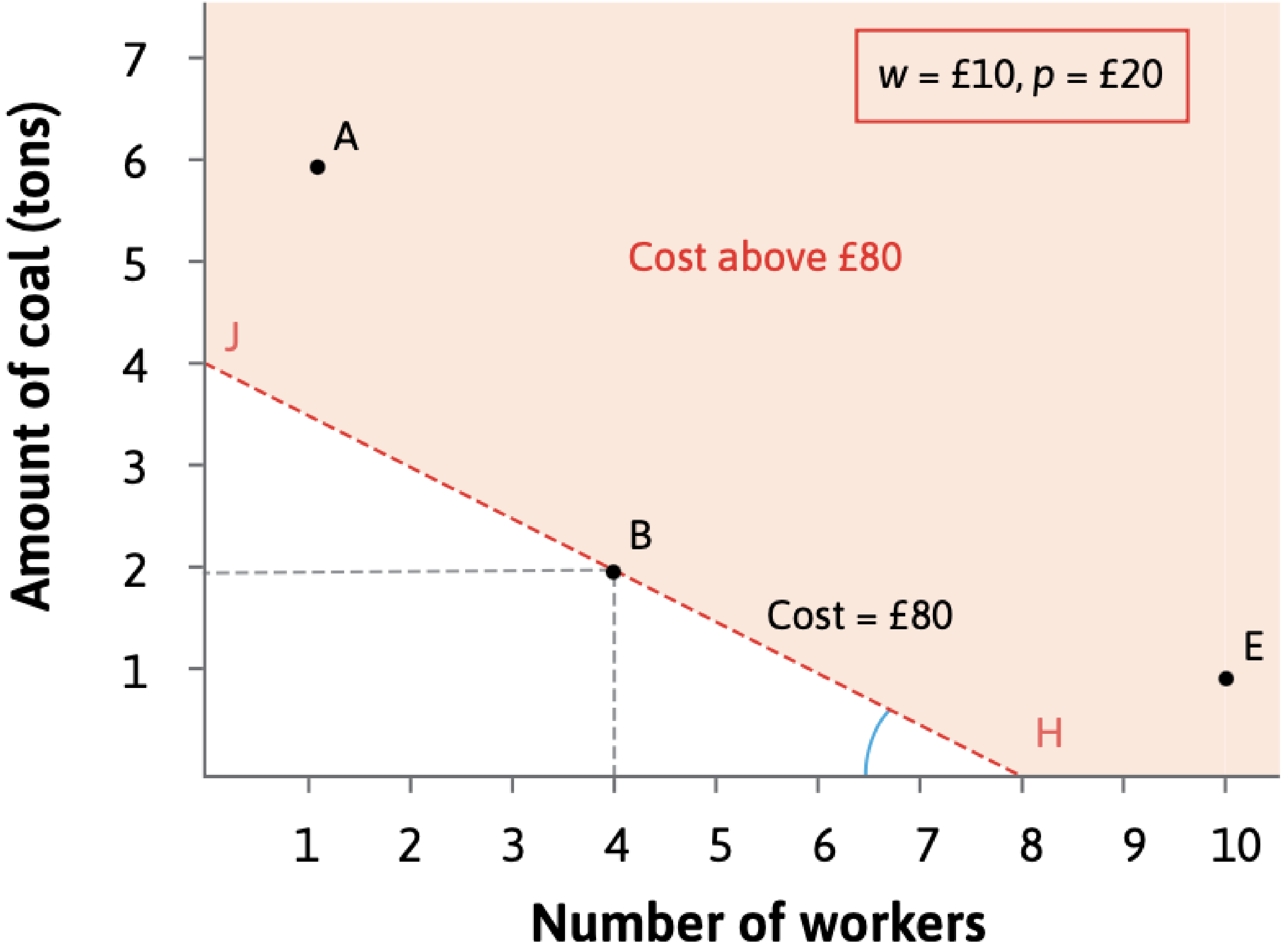


Which technology?

- Isocost lines join all the combinations of workers and coal that cost the same amount
- Slope of isocost lines = -w/p
- Slope depends on <u>relative price</u>



Which technology?

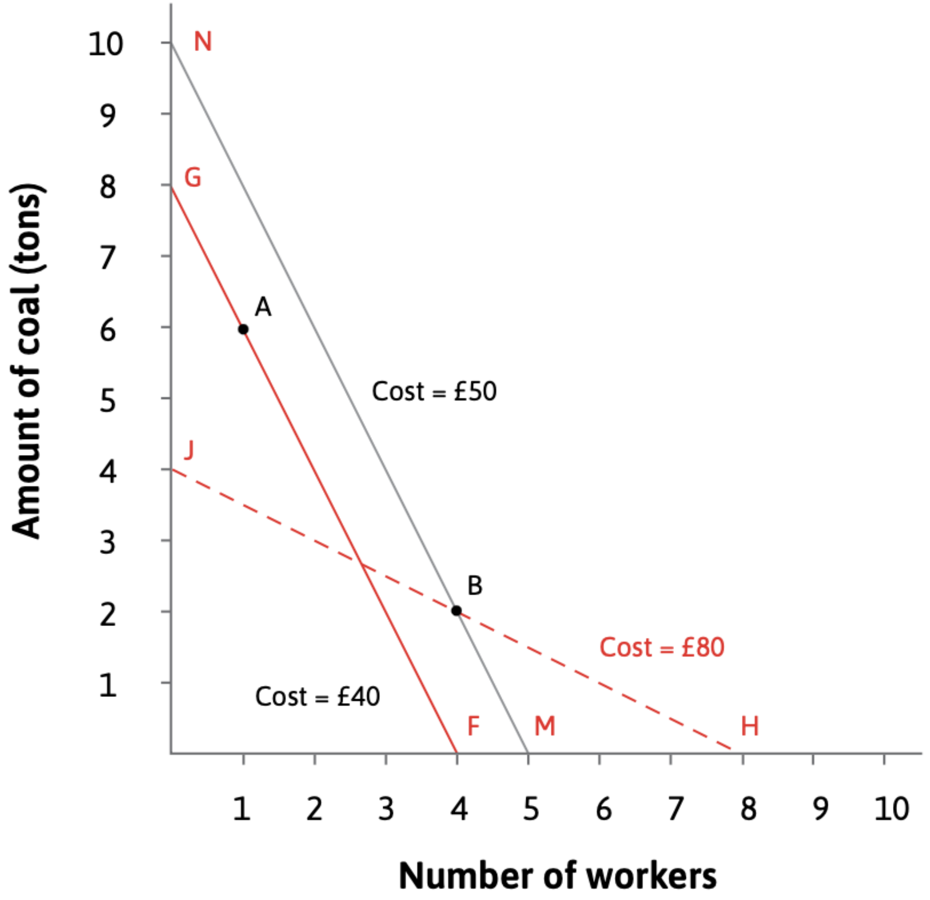


Technology	Number of workers	Coal required (tons)	Total cost (£)
A	1	6	130
B	4	2	80
E	10	1	120
Wage £10, coal price £20 per ton			

Figure 2.8 The costs of using different technologies to produce 100 metres of cloth.

A change in relative prices

- Suppose the price of coal falls to £5, the wage remains at £10 and the price of cloth stays the same.



Examples of creative destruction

- In groups, come up with recent examples of creative destruction (for example, finding case studies in the newspaper).
- OR you can use an AI assistant to generate recent examples of creative destruction, prompt "*List five recent examples of creative destruction in various industries, explaining how new technologies or business models have replaced older ones*" instead.

Exercise 2.5: Isocost lines

Suppose the wage is £10 but the price of coal (per ton) is only £5.

- What is the relative price of labour?
- On a diagram with the number of workers on the horizontal axis and tons of coal on the vertical axis, draw the isocost line that shows all combinations of inputs that cost £60. (Hint: Find and connect both endpoints of this line.)
- On the same diagram, draw the isocost lines corresponding to costs of £30 and £90. How does the set of isocost lines for these input prices compare to the ones for w = 10 and p = 20?

Exercise 2.5: Isocost lines (Answers)